Annexure3b- Complete filing
INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE: AI-Powered Smart Podium with Integrated Cleaning & Mobile Detection for Enhanced Classroom Experience

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

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Signature (Mandatory)	

3. DESCRIPTION OF THE INVENTION:

The Smart Podium is an AI-powered lecture stand that enhances classroom teaching with biometric authentication, smart connectivity, mobile detection, and automated cleaning. It features built-in HDMI, AUX, and USB mic support for seamless device integration and a real-time mobile detection system to prevent distractions. A robotic vacuum cleaner ensures classroom hygiene, while an emergency dashboard provides instant alerts for security and staff, making it a comprehensive smart solution for modern classrooms.

A. PROBLEM ADDRESSED BY THE INVENTION: Traditional podiums lack automation, smart authentication, security alerts, and mobile phone detection. Teachers face difficulties such as:

a. Manually handling lecture materials & device connections.
b. Unauthorized mobile phone usage disrupting classrooms.
c. Lack of integrated security features in case of emergencies.
d. Poor audio clarity due to classroom noise.
e. Messy classrooms due to poor cleaning and dust accumulation.

B. OBJECTIVE OF THE INVENTION:

a. To develop a smart, AI-powered podium with biometric authentication, emergency response, and automated cleaning.
b. To integrate real-time RF-based mobile phone detection to prevent distractions during class.
c. To enhance lecture audio clarity with a USB-dongle-based wireless mic system for teachers.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY:

Sr. No.	Patent I'd	Abstract	Research Gap	Novelty
1.	202311045887	Smart podium with device connectivity & automation	Lacks smart cleaning, mobile phone detection, and mic integration	Built-in robotic vacuum cleaner, RF-based mobile detection, & USB mic system

D. DETAILED DESCRIPTION:

The Smart Podium is an advanced AI-powered lecture stand designed to enhance classroom teaching by integrating biometric authentication, emergency response systems, automation, and real-time device monitoring. The podium features biometric authentication through fingerprint or facial recognition, allowing teachers to log in securely. The system is programmed to automatically log out after 55 minutes (the standard class duration) to prevent unauthorized access, while also providing an option for manual logout if needed.

For seamless connectivity, the podium is equipped with two power sockets for charging laptops or devices, along with built-in pull-out HDMI and AUX cables that allow easy connection to external projectors and speakers. Additionally, it features a USB-dongle-based 2.5GHz wireless microphone, which automatically connects to the classroom speaker system, enhancing the teacher's voice clarity without requiring additional setup.

To address emergency situations, the podium incorporates a dedicated emergency dashboard with a single-touch alert system. Pressing the emergency button immediately notifies security personnel, ambulance services, the nearest staff room, and the exam control room (with a dynamic location update based on the ongoing exam schedule).

The podium also offers smart storage options, with dedicated slots on all sides to store essential teaching tools such as markers, a duster, and a water bottle.

One of the most innovative features is the automated cleaning system, which includes a built-in robotic vacuum cleaner housed at the podium base. The vacuum cleaner is scheduled to activate at 8 PM daily, cleaning the classroom floor before returning to its docking station, where it locks itself inside the podium. A special maintenance unlock feature allows authorized personnel to access the vacuum cleaner for servicing and cleaning.

To combat unauthorized mobile phone usage, the podium is equipped with an RF-based mobile phone detection system that scans frequencies of 3 GHz or lower to detect active mobile devices. A heatmap display on the podium screen visually represents the location of mobile activity in the classroom. Furthermore, a transparent film overlay on the screen helps the teacher identify the exact seating position of students using mobile phones, allowing for real-time monitoring and control.

Overall, the Smart Podium offers a revolutionary approach to classroom teaching by integrating cutting-edge automation, security, and smart connectivity, ensuring an efficient, distraction-free, and technologically advanced learning environment.

E. RESULTS AND ADVANTAGES:

1. Enhances classroom efficiency by automating lecture management.
2. Provides seamless device connectivity through HDMI, AUX, and USB mic integration.
3. Improves classroom security with a built-in emergency alert system.
4. Reduces distractions by detecting unauthorized mobile usage.
5. Ensures better classroom hygiene through automated cleaning.

F. EXPANSION:

1. Cloud connectivity for storing teacher preferences and schedules.
2. AI-powered attendance tracking through face recognition.

G. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION: No, Not yet decided

G. EXISTING DATA: No existing comparative data available yet. However, the invention improves upon patent 202311045887 by integrating smart cleaning, mobile phone detection, and an enhanced mic system, making it a more advanced and efficient solution.

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?	YES ()	NO (*)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?	YES ()	NO (*)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?	YES ()	NO (*)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ()	NO (*)
E. Name of Regulatory body or any other approvals if required.	YES ()	NO (*)

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us.: Not Shared

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity):NO

7. Potential Chances of Commercialization.

1. Educational institutions (Schools, Colleges, Universities).

2. Corporate training centres.
3. Government training halls.

8. List of companies which can be contacted for commercialization along with the website link.

9. Any basic patent which has been used and we need to pay royalty to them.

- Any existing patents used? Yes (Patent ID: 202311045887).
- Does it require royalty payments? Not yet known.

10**. FILING OPTIONS:** Provisional Patent

11. **KEYWORDS:** Smart Podium, AI Podium, Smart Lecture Stand, Automated Teaching Stand, Mobile Detection in Class, Classroom Security System, Smart Classroom Automation.

Welcome Dr. Monica Gulati Sign out

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सत्यमेव जयते

INTELLECTUAL PROPERTY INDIA
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G.A.R.6
[See Rule 22(1)]
RECEIPT

Docket No 94413 **Date/Time 2025/09/09 14:57:02**

To
Dr. Monica Gulati

UserId: Registrar

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CBR Detail:

Sr. No.	App. Number	Ref. No./Application No.	Amount Paid	C.B.R. No.	Form Name	Remarks
1	E-106/12156/2025/DEL	202511085426	0	-----	FORM28	
2	E-106/12158/2025/DEL	202511085429	0	-----	FORM28	----
3	E-106/12153/2025/DEL	202511085431	0	-----	FORM28	----
4	E-106/12159/2025/DEL	202511085433	0	-----	FORM28	----
5	E-106/12157/2025/DEL	202511085432	0	-----	FORM28	----
6	E-106/12151/2025/DEL	202511085434	0	-----	FORM28	----
7	E-106/12160/2025/DEL	202511085435	0	-----	FORM28	----
8	E-106/12155/2025/DEL	202511085436	0	-----	FORM28	----
9	E-106/12154/2025/DEL	202511085428	0	-----	FORM28	----
10	E-106/12162/2025/DEL	202511085437	0	-----	FORM28	----
11	E-106/12152/2025/DEL	202511085430	0	-----	FORM28	----
12	E-106/12161/2025/DEL	202511085427	0	-----	FORM28	----

13	202511085433	TEMP/E-1/58921/2025-DEL	1600	46731	FORM 1	A COMPOSITION FOR SUBLINGUAL ADMINISTRATION OF ANTI-INFLAMMATORY BOTANICALS AND A METHOD FOR MUCOSAL
14	202511085432	TEMP/E-1/58923/2025-DEL	1600	46731	FORM 1	WOMEN'S WELLNESS GUMMIES WITH SHATAVARI, CRANBERRY EXTRACTS, VITAMIN D3, AND FORMULATION METHOD
15	202511085434	TEMP/E-1/58924/2025-DEL	1600	46731	FORM 1	A METHOD FOR OPERATING A SMART PODIUM IN A CLASSROOM ENVIRONMENT
16	202511085435	TEMP/E-1/58927/2025-DEL	1600	46731	FORM 1	A SYSTEM FOR AUTONOMOUS AGRICULTURAL PROTECTION AND MONITORING
17	202511085436	TEMP/E-1/58928/2025-DEL	1600	46731	FORM 1	MULTIFUNCTIONAL NUTRACEUTICAL CAPSULE FOR HOLISTIC AGING DEFENSE TARGETING CELLULAR HEALTH AND PREPA
18	202511085428	TEMP/E-1/58930/2025-DEL	1600	46731	FORM 1	A SYSTEM FOR PERSONALIZED MENTAL HEALTH MANAGEMENT
19	202511085437	TEMP/E-1/58931/2025-DEL	1600	46731	FORM 1	CHEWABLE COMPOSITION FOR BONE DENSITY AND JOINT FUNCTION SUPPORT WITH A METHOD FOR PREPARATION
20	202511085430	TEMP/E-1/58938/2025-DEL	1600	46731	FORM 1	A SMART CART SYSTEM FOR RETAIL SHOPPING
21	202511085427	TEMP/E-1/58943/2025-DEL	1600	46731	FORM 1	A PERSONAL EMERGENCY DETECTION AND RESPONSE SYSTEM
22	202511085426	TEMP/E-1/58675/2025-DEL	1600	46731	FORM 1	A METHOD FOR CONTROLLING ACCESS USING A SMART LOCK SYSTEM
23	202511085429	TEMP/E-1/58689/2025-DEL	1600	46731	FORM 1	A METHOD FOR REGULATING SEAT TEMPERATURE IN A VEHICLE
24	202511085431	TEMP/E-1/58727/2025-DEL	1600	46731	FORM 1	A METHOD FOR PROTECTING MOVABLE AND IMMOVABLE ASSETS DURING FLOODING

TransactionID	Payment Mode	Challan Identification Number	Amount Paid	Head of A/C No
N-0001746927	Online Bank Transfer	0909250031258	19200.00	1475001020000001

Total Amount : ₹ 19200.00

Amount in Words: Rupees Nineteen Thousand Two Hundred Only

Received from Dr. Monica Gulati the sum of ₹ 19200.00 on account of Payment of fee for above mentioned Application/Forms.

* This is a computer generated receipt, hence no signature required.

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